

Variable Importance

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Introduction

Variable importance ranks features by contribution to model accuracy. It is the most common first-pass interpretability tool for black-box models. Tree ensembles offer two native measures: impurity-based (mean decrease in Gini or variance) and permutation-based (drop in accuracy when the feature is shuffled).

Prerequisites

Decision trees; random forests; cross-validation.

Theory

Impurity importance: sum the impurity decrease over all splits using feature j . Biased toward high-cardinality features and correlated features.

Permutation importance: shuffle feature j in held-out data; measure the drop in performance. Unbiased for correlated features as a group but can split importance between correlated pairs.

Conditional permutation and SHAP values handle correlation more rigorously (at higher cost).

Assumptions

Sufficient held-out data for permutation; features correctly treated as continuous or categorical.

R Implementation

```
library(ranger); library(vip)

set.seed(2026)
n <- 500
df <- data.frame(
  x1 = rnorm(n), x2 = rnorm(n), x3 = rnorm(n),
  x4 = rnorm(n), x5 = rnorm(n),
  y = sin(rnorm(n)) + 0.3 * rnorm(n)^2
)

rf <- ranger(y ~ ., data = df, num.trees = 500,
  importance = "permutation")
imp <- sort(rf$variable.importance, decreasing = TRUE)
print(round(imp, 3))

vip(rf, num_features = 5) +
  ggplot2::labs(title = "Permutation importance (ranger RF)")
```

Output & Results

Sorted importance scores; the top feature dominates, others are close to zero on this mostly-noise data.

Interpretation

“Permutation importance identified x_1 as the dominant predictor (12 % RMSE increase when shuffled); other features contributed negligibly, consistent with the data-generating process.”

Practical Tips

- Use permutation importance when correctness matters; impurity importance is cheap but biased.
- Drop correlated features or use grouped importance if groups of correlated variables share the signal.
- Importance alone does not reveal direction of effect; pair with partial dependence or SHAP.
- Normalise importance by the baseline model error to express as a relative increase.
- For deep learning, use integrated gradients or SHAP; RF-style permutation is not standard.

Reporting

When reporting variable importance, always state which flavour was used (impurity, permutation, drop-column, or SHAP), the loss against which importance is measured, and whether the held-out test set or the training data drove the calculation. Permutation importance computed on training data overstates the apparent role of variables that the model overfits to, so the test-set version is the more defensible default. In a manuscript, include a small bar chart or table for the top features and note explicitly that importance is a measure of model reliance, not of causal effect or biological mechanism. Where features are correlated, mention that shared signal can split importance across columns and that grouped-permutation summaries may be more interpretable.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis